Stress Detection with Machine Learning

Stress, anxiety, and depression are threatening the mental health of people. Every person has a reason for having a stressful life. People often share their feelings on social media platforms like on Instagram in the form of posts and stories, and on Reddit in the form of asking for suggestions about their life on subreddits. In the past few years, many content creators have come forward to create content to help people with their mental health. Many organizations can use stress detection to find which social media users are stressed

The dataset I am using for this task contains data posted on subreddits related to mental health. This dataset contains various mental health problems shared by people about their life. Fortunately, this dataset is labelled as 0 and 1, where 0 indicates no stress and 1 indicates stress. So in the section below, I will take you through the task of stress detection in social media posts using Python.

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import LabelEncoder

from sklearn.naive_bayes import BernoulliNB
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier, VotingClassifier

from xgboost import XGBClassifier
```

```
import pandas as pd
import numpy as np
data = pd.read_csv("/content/drive/MyDrive/Stress-detection-using-NLP/stress.csv")
data.head()
```

	subreddit	post_id	sentence_range	text	id	label	confidence	social_timestamp	social_karma	synt
0	ptsd	8601tu	(15, 20)	He said he had not felt that way before, sugge...	33181	1	0.8	1521614353	5	1
1	assistance	8lbrx9	(0, 5)	Hey there r/assistance, Not sure if this is th...	2606	0	1.0	1527009817	4	9
2	ptsd	9ch1zh	(15, 20)	My mom then hit me with the newspaper and it s...	38816	1	0.8	1535935605	2	7
3	relationships	7rorpp	[5, 10]	until i met my new boyfriend, he is amazing, h...	239	1	0.6	1516429555	0	2
4	survivorsofabuse	9p2gbc	[0, 5]	October is Domestic Violence Awareness Month a...	1421	1	0.8	1539809005	24	7

5 rows × 116 columns

```
data.shape
```

```
(2838, 116)
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
pd.options.display.max_rows = None
print(data.isnull().sum())
```

```
lex_liwc_discrep            0
lex_liwc_tentat             0
lex_liwc_certain            0
lex_liwc_differ             0
lex_liwc_percept            0
lex_liwc_see                0
lex_liwc_hear               0
lex_liwc_feel               0
lex_liwc_bio                0
lex_liwc_body               0
lex_liwc_health             0
lex_liwc_sexual             0
lex_liwc_ingest             0
lex_liwc_drives             0
lex_liwc_affiliation        0
lex_liwc_achieve            0
lex_liwc_power              0
lex_liwc_reward             0
lex_liwc_risk               0
lex_liwc_focuspast          0
lex_liwc_focuspresent       0
lex_liwc_focusfuture        0
lex_liwc_relativ            0
lex_liwc_motion             0
lex_liwc_space              0
lex_liwc_time               0
lex_liwc_work               0
lex_liwc_leisure            0
lex_liwc_home               0
lex_liwc_money              0
lex_liwc_relig              0
lex_liwc_death              0
lex_liwc_informal           0
lex_liwc_swear              0
lex_liwc_netspeak           0
lex_liwc_assent             0
lex_liwc_nonflu             0
lex_liwc_filler             0
lex_liwc_AllPunc            0
lex_liwc_Period             0
lex_liwc_Comma              0
lex_liwc_Colon              0
lex_liwc_SemiC              0
lex_liwc_QMark              0
lex_liwc_Exclam             0
lex_liwc_Dash               0
lex_liwc_Quote              0
lex_liwc_Apostro            0
lex_liwc_Parenth            0
lex_liwc_OtherP             0
lex_dal_max_pleasantness    0
lex_dal_max_activation      0
lex_dal_max_imagery         0
lex_dal_min_pleasantness    0
lex_dal_min_activation      0
lex_dal_min_imagery         0
lex_dal_avg_activation      0
lex_dal_avg_imagery         0
```

Luckily, there are no missing values found in the dataset

```
import nltk
import re
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
stemmer = nltk.SnowballStemmer("english")
from nltk.corpus import stopwords
import string
stopword=set(stopwords.words('english'))
```

```
import matplotlib.pyplot as plt
from wordcloud import WordCloud, STOPWORDS, ImageColorGenerator
text = " ".join(i for i in data.text)
stopwords = set(STOPWORDS)
wordcloud = WordCloud(stopwords=stopwords, font_path="/usr/share/fonts/truetype/liberation/LiberationSans-Regula
plt.figure( figsize=(15,10))
plt.imshow(wordcloud, interpolation="bilinear")
```

```
plt.axis("off")
plt.show()
```

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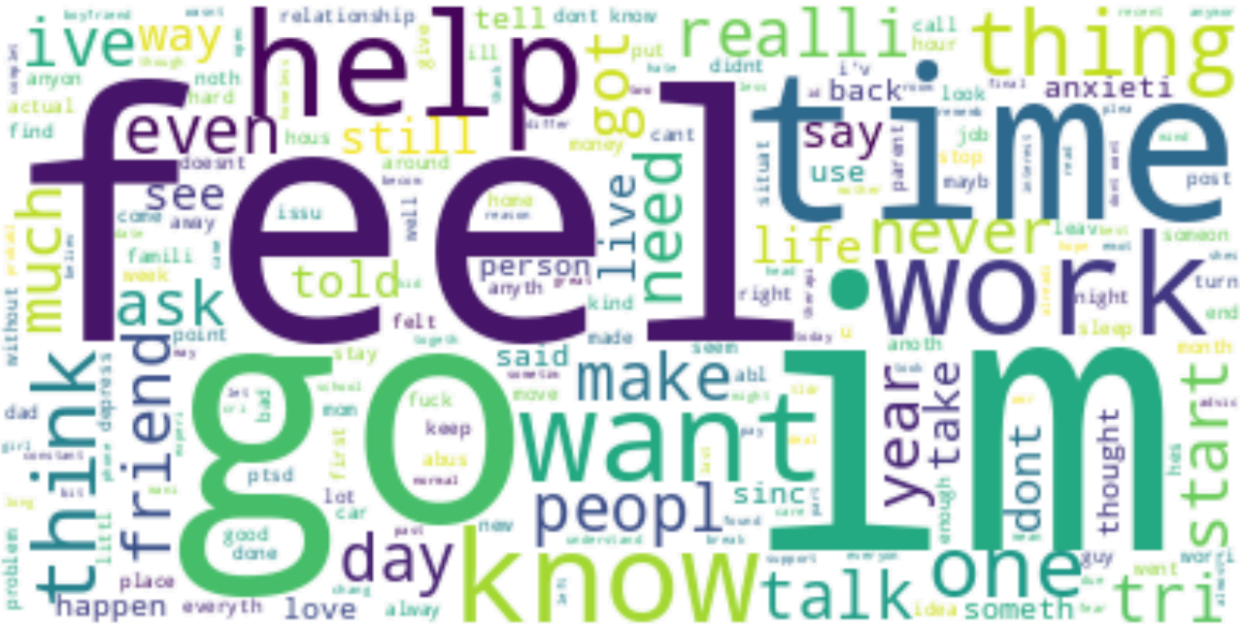
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```
    pipeline.fit(xtrain, ytrain)
    ypred = pipeline.predict(xtest)

    acc = accuracy_score(ytest, ypred)
    print(f"{name} Accuracy: {acc:.4f}")



voting_model = VotingClassifier(
    estimators=[
        ('nb', BernoulliNB()),
        ('rf', RandomForestClassifier(n_estimators=100)),
        ('svm', SVC(probability=True))
    ],
    voting='soft'
)

voting_pipeline = Pipeline([
    ('vectorizer', TfidfVectorizer()),
    ('model', voting_model)
])

voting_pipeline.fit(xtrain, ytrain)
ypred_vote = voting_pipeline.predict(xtest)

print("\n---- Ensemble Model ----\n")
print("Voting Classifier Accuracy:", accuracy_score(ytest, ypred_vote))


sample = ["I feel overwhelmed and stressed with work"]

pred = voting_pipeline.predict(sample)
label = le.inverse_transform(pred)

print("\nSample Prediction:", label[0])
```

```
Naive Bayes Accuracy: 0.7406
SVM Accuracy: 0.7453
Random Forest Accuracy: 0.7101
/usr/local/lib/python3.12/dist-packages/xgboost/training.py:200: UserWarning: [18:49:39] WARNING: /__w/xgboost/xg
Parameters: { "use_label_encoder" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
XGBoost Accuracy: 0.7230

---- Ensemble Model ----

Voting Classifier Accuracy: 0.7535211267605634

Sample Prediction: Stress
```

Start coding or generate with AI.